

Real-time 3D-Lidar, MMW Radar and GPS/IMU fusion based vehicle detection and tracking in unstructured environment*

Ning Li, Caixia Lu, Xuewei Yu, Xueyan Liu, Bo Su

Abstract—To solve the problem of unmanned ground vehicle leader-follower formation transportation in unstructured environment, we propose a novel target detection and tracking method based on multi-sensor fusion perception. Combined with 3D-Lidar, millimeter wave Radar and GPS/IMU, the proposed method can achieve stable target detection and continuous tracking of both static and dynamic vehicles. First, 3D-Lidar is used to detect the geometric model of the leader vehicle to complete the initialization of tracking target and it can also be assisted for target tracking. Then during the movement, the dynamic leader is mainly tracked through millimeter wave Radar as this sensor can keep tracking the same target with a constant index and effectively distinguish dynamic vehicle from other static obstacles according to relative speed estimation. In addition, by using GPS/IMU based integrated navigation, the movement trend of the leader can be derived according to the echo vehicle pose information and the relative position relationship. This is helpful to reduce the region of interest for target tracking and improve the real-time performance. In different unstructured environments, we perform the leader-follower formation transportation experiments for hundreds of kilometers. In rough terrain, the maximum tracking speed can still reach 40km/h and the maximum tracking distance can be up to 100 meters. Experiments show that the proposed method is suitable for vehicle target detection and tracking in unstructured environment. It has good robustness and high real-time performance with an average processing frame rate of 20Hz. The proposed method can be used for the formation transportation of unmanned ground vehicles to reduce labor costs.

Key words: vehicle detection and tracking, unstructured environment, multi-sensor fusion, unmanned ground vehicle

I. INTRODUCTION

The leader-follower formation transportation is a useful semi-autonomous navigation control method for unmanned ground vehicle (UGV) [1]. By detecting and tracking the leader vehicle through sensors, the formation transportation system can be achieved without retrofitting the leader. The detection and continuous stable tracking of the leader vehicle in unstructured environment is the focus in our study. In the structured scene, the road is flat without occlusion and there are traffic rules, the leader and follower drive in the same lane, so the target detection and continuous tracking are relatively simple. Compared with this, vehicle detection and tracking in unstructured environment is much hard. Problems are

described as follows: 1) The road is bumpy in rough terrain. It is easy to cause loss of tracking target. 2) For sand or dirt road environment, especially in windy conditions, during the movement, dust caused by the leader will adversely affect targets detection and tracking. 3) Weather factor. For weather conditions such as light rain, wind and fog, the anti-interference perception ability of sensors needs to be considered. 4) Occlusion. When the leader drives with trees on road sides, partial occlusion by branches and leaves may occur. Meanwhile the field of view of sensors need to be considered to prevent loss of tracking target during turning. 5) Vehicle interaction. It is necessary to deal with the interference of other vehicles on the road to prevent mis-tracking.

A. Related Work

For the detection and tracking of the leader vehicle, sensors that can be used commonly include camera (color camera and infrared camera), 3D-Lidar, millimeter wave Radar (MMW Radar) and so on. All these above sensors have been studied by researchers to solve the problem.

For camera, deep learning based vehicle detection and tracking technology has been extensively and deeply researched in recent years. Algorithms such as YOLO [2-4] and SSD [5] are excellent methods for vehicle detection in the field of computer vision and image processing. For continuous target tracking, paper [6] proposes a Siamese region proposal network by end-to-end training. These methods meet real-time requirements based on GPU graphics computing and have good performance in data sets testing like KITTI with structured environment. However, on one hand, in unstructured scenes, roads are complex and image captured by camera are often unstable. Image jitter caused by rough terrain, the change of light, partial occlusion of surrounding obstacles will seriously affect the detection and tracking result. On the other hand, due to the limited view field of cameras, when the leader turns, the target is easy to be beyond the field of view in order to cause the loss of it. Meanwhile the location of the target in the image frame needs to be transformed to the vehicle's local frame for motion planning by perspective transformation. There is inevitable conversion deviation in the transformation process so that it is difficult to provide an accurate and stable location for the follower.

For 3D-Lidar, the leader's location can be obtained directly from the point cloud space without perspective transformation compared with camera and the horizontal field of view is 360° without blind angle. Paper [7] proposes a method by using both dynamic and geometric properties of the vehicle with single Bayes filter. This method has been validated at the Urban Grand Challenge. In addition, deep learning technology which is widely used in image processing are also applied for 3D target detection. In paper [8], a two-stage convolutional neural

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network is used for 3D detection of vehicles. These methods are mainly used to solve tracking problems in urban environment. In the case of bumpy roads with dust in unstructured environment, the performance of these methods will decline sharply as 3D-Lidar has a poor anti-interference ability for dust. Besides, point clouds become sparse with increasing distance in order that the detection accuracy rate drops rapidly in the distance.

MMW Radar has a strong anti-interference ability against dust, rain, fog and snow. Currently, MMW Radar is widely used in automotive safety and adaptive cruise [9]. However, it cannot distinguish the static vehicle from surrounding obstacles which is important for the initialization of target tracking. When the leader starts statically, it needs other sensors to provide the initial tracking target. In addition, when turning, due to the limited field of view, it is also easy to lost tracking target.

B. Our Work

Each sensor has its advantages and disadvantages. Thus, for complex unstructured environment, we use a multi-sensor fusion method to achieve the vehicle detection and continuous tracking. The fusion processing makes full use of characteristics of each sensor to adapt to the complex and changeable unstructured environment. Sensors used in the paper include 3D-Lidar, MMW Radar and GPS/IMU. First, 3D-Lidar is used to solve the problem of tracking target initialization. Geometric features of vehicle model in point cloud space are used for initial target detection. As the horizontal field of view of 3D-Lidar is 360°, it can effectively prevent target loss when the leader turns. And in close range, it can also be assisted for target tracking. Then MMW Radar is mainly used for dynamic target tracking during the movement. It can track the same target with a constant index continuously and steadily. Besides, according to the feedback relative speed information, it can effectively distinguish the dynamic vehicle from surrounding static obstacles. Finally, for extreme situations such as sharp turning and slope, we use the echo pose information provided by GPS/IMU to retain the absolute position of the leader when the target cannot be detected by sensors. It is useful to re-track the target during short-term target loss. Meanwhile, the analysis of the leader's movement trend based on echo pose information and the relative position relationship is helpful to reduce the region of interest to ensure the real-time performance.

The reminder of the paper is organized as follow. Section II is system architecture. Section III is 3D-Lidar based vehicle detection and auxiliary tracking. Section IV is MMW Radar based vehicle tracking. Section V is the experiment and Section VI is the conclusion.

II. SYSTEM ARCHITECTURE

A. Multi-sensor Fusion Hardware System

For formation transportation of UGVs, stable detection and continuous tracking of the leader is the key for the technology to be applied in practice. We use multi-sensor fusion technology to realize detection and tracking in unstructured environment to overcome various complex situations. The hardware sensors include 3D-Lidar, MMW Radar and Global Positioning System (GPS) / Inertial Measurement Unit (IMU). Pandar40 is selected as 3D-Lidar

with 360° horizontal field of view. Delphi ESR is selected as MMW Radar with $\pm 45^\circ$ horizontal field of view and the output is 64 tracking targets with relative position and relative speed. During continuous tracking, the index of each target remains unchanged until it disappears. XW-GI7660 is selected as the integrated navigation device. We can get the echo pose information from the integrated result of INS/GPS and the output includes yaw, pitch, roll, global positioning, vehicle speed information and so on. Fig.1 shows the related sensors.



Fig. 1. Multi-sensor fusion hardware system.

B. Software System Overview

The installation position of each sensor is different. The 3D-Lidar is mounted on the roof of UGV, the MMW Radar is installed on the front side, and the IMU is installed on the center of the rear axle of UGV. All sensor data should be unified to the same coordinate system in order to facilitate data fusion processing. The unified coordinate system we select is the local coordinate system of the vehicle body defined as F_v . The center of the rear axis of the vehicle projected onto the ground is selected as the coordinate origin. The heading direction of the vehicle is the positive direction of y axis, the right of the vehicle is the positive direction of the x axis and the z direction is upward pointing to the sky. In addition, during the continuous tracking processing, the detected position of the leader in the local frame of F_v needs to be transformed to a global frame to form a continuous trajectory for movement trend analysis of the leader. The China Geodetic Coordinate System 2000 is used as the global frame defined as F_G . This information can also be used to associate the historical tracking trajectory with the current result through coordinate transformation in order to extract the region of interest.

As shown in Fig.2, first, 3D-Lidar is used to perform target detection to provide an initial tracking target. When the leader drives during a certain distance and the tracking target is stable by 3D-Lidar, we match the same target generated by MMW Radar. During the movement, the continuous dynamic tracking result mainly depend on MMW Radar. When the tracking target is lost by MMW Radar due to turning or other factors, the auxiliary tracking by 3D-Lidar is temporarily used. Meanwhile the tracking result of MMW Radar is tried to re-match with that based on 3D-Lidar and then the target tracking based on MMW Radar continues. The black dotted lines indicate that the 3D-Lidar based auxiliary tracking is not activated when tracking based on MMW Radar is valid.

In the continuous tracking processing, the historical tracking trajectory of the leader can be associated by echo pose information provided by GPS/IMU. The trajectory can be used to analyze the movement trend of the leader through Kalman predictor. This processing involves the mutual transformation of F_v and F_G . This is helpful to reduce the region of interest and improve the real-time performance. In addition, when the tracking target is partially or completely occluded by trees or walls, the global position of the last valid tracking target can be transformed into the current frame to

ensure that the follower can drive to the lost position and restart target detection for tracking initialization.

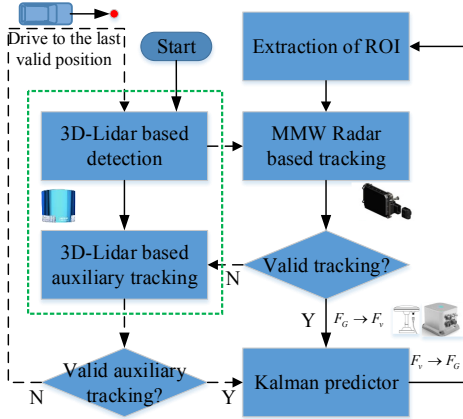


Fig.2 The proposed algorithm flowchart.

III. 3D-LIDAR BASED VEHICLE DETECTION AND AUXILIARY TRACKING

Vehicle detection result is used for tracking target initialization, so it is the prerequisites of continue tracking. The geometric model of the leader vehicle can be described by point clouds produced by 3D-Lidar. Although target detection is mainly used for tracking initialization when the leader is static, we still consider the problem of real-time performance. The reason is that target detection result can also be applied to target re-initialization when the tracking target is lost during the movement. Besides, Auxiliary tracking based on 3D-Lidar will work when tracking by MMW Radar fails. The processing flow is shown in Fig. 3.

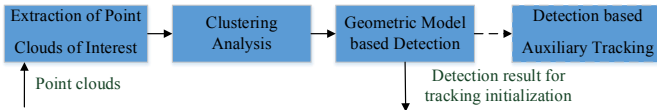


Fig.3 3D-Lidar based vehicle detection and auxiliary tracking.

A. Extraction of Point Clouds of Interest

In point cloud space, most point clouds are scanned onto the ground, but they are of useless for our research. So first, obstacle point clouds are extracted as point clouds of interest, because point clouds scanned to the leader vehicle are also among them. This dealing is helpful to reduce the amount of point clouds. We use the method provided in [10] to extract obstacle point clouds. According to this method, all point clouds are vertically projected to the 2D grid map. The height differences of point clouds projected into the same grid can be used to judge whether it is an obstacle grid or not. Point clouds projected into the obstacle grid are labeled as obstacle point clouds.

As obstacle point clouds are perpendicular to the ground plane, obstacle detection based on height difference information between neighborhoods is effective. In rough terrain, to eliminate the influence of road undulations, we use pitch and roll information to correct the coordinate of point clouds as shown in (1).

$$P_c = R(\gamma, \beta, \alpha = 0) \cdot P_v = R_x(\gamma) \cdot R_y(\beta) \cdot P_v \quad (1)$$

Where P_v represents the source point cloud in F_v , P_c represents the corrected point cloud, γ represents the pitch angle, β represents the roll angle, α represents the yaw angle and R represents the rotation matrix.

Then the maximum height difference $\Delta_{x,y}$ in each grid cell can be calculated by (2).

$$\Delta_{x,y} = \max \{ \|z_{x,y} - G_{x,y}^{ref}\| \mid z_{x,y} \in Z_{x,y} \} \quad (2)$$

Where $z_{x,y}$ is the height value of one point cloud in the set $Z_{x,y}$ with the coordinate of (x,y) in the grid map. $G_{x,y}^{ref}$ is the reference height of the surrounding ground plane calculated by (3).

$$G_{x,y}^{ref} = \min \{ z_{xx,yy} \mid (xx,yy) \in N - \{n\}(x,y) \} \quad (3)$$

Where $N - \{n\}(x,y)$ represents the n -neighborhoods of (x,y) .

The grid cell with $\Delta_{x,y} > \Delta_T$ is labeled as obstacle occupation and the associated point clouds are extracted as ones of interest. The extracted obstacle point clouds are colored red in Fig.4 which include those scanned to the leader.

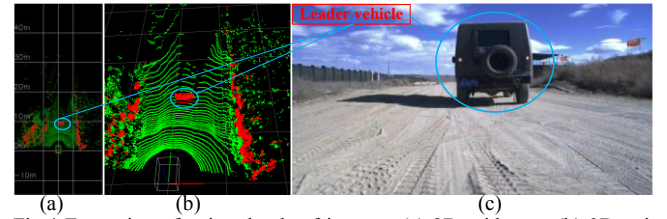


Fig.4 Extraction of point clouds of interest. (a) 2D grid map, (b) 3D point cloud space, (c) image. Point clouds colored red are ones of obstacles.

There are irrelevant point clouds such as buildings, trees and dynamic pedestrians among obstacle point clouds. These irrelevant point clouds are so large as to increase time complexity. We can eliminate some irrelevant obstacles based on the analysis of each obstacle's size. The binary map in Fig.5(a) is derived from the 2D grid map in Fig.4(a) according to the occupation of obstacle, where the white area represents obstacle occupation. Connected component analysis of the binary map is performed to obtain obstacle domains and then the minimum bounding box of each obstacle domain is calculated. The size of each box is used as constraints to further eliminate irrelevant point clouds belonging to very large or small obstacles and point clouds colored purple in Fig.5(c) represents ones after filtering corresponding to those shown in Fig.5(b).

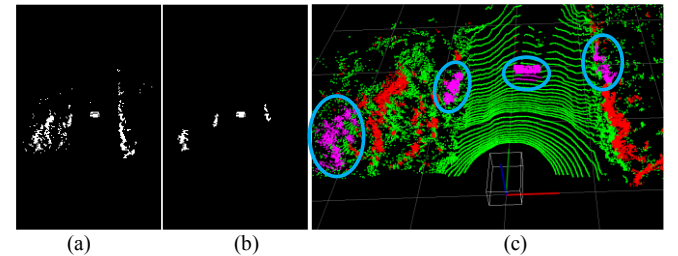


Fig.5 Filtering of point clouds. (a) the binary map of obstacles, (b) the binary map after filtering, (c) point clouds after filtering which are colored purple.

B. Clustering Analysis

Before clustering, down-sampling is used to reduce the amount of point clouds. Here we use voxel grid method to achieve down-sampling and the voxel after down-sampling is represented by the centroid of corresponding point clouds. The voxel size defined in this paper is 0.1m^3 .

Then we use the K-D tree method to perform point cloud clustering in 3D space and the Euclidean distance between point clouds is used as the criteria of clustering shown in (4).

$$\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2} \leq \Delta_R \quad (4)$$

Where (x_i, y_i, z_i) and (x_j, y_j, z_j) are two point clouds and the threshold distance is defined as Δ_R . After clustering, point clouds are divided into different clustering blocks.

C. Geometric Model based Vehicle Detection

For each clustering block, the minimum bounding box is useful to describe the overall feature of the vehicle. In the actual processing, we need to judge the relative parking direction of the detected vehicle in point cloud space as the leader may not stop right in front of the follower especially during turning. The length and width of the leader can be obtained by analyzing the bounding box. Since we transform point clouds to the vehicle frame, the height direction can be determined to be perpendicular to the ground and the height information can be directly obtained through statistics in the height direction. Then we set the height of all point clouds in each clustering block to be zero, and only analyze the horizontal distribution characteristics of these points to calculate feature vectors in order to get the length and width of the block.

Here the principle component analysis (PCA) is used to calculate feature vectors of point clouds. First, calculate the centroid and the covariance matrix of each clustering block defined as $(\bar{x}, \bar{y}, \bar{z})$ and C in (5) and (6).

$$(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{\sum_{i=1}^n x_i}{n}, \frac{\sum_{i=1}^n y_i}{n}, \frac{\sum_{i=1}^n z_i}{n} \right) \quad (5)$$

$$C = \begin{bmatrix} c_{x,x} & c_{x,y} & c_{x,z} \\ c_{y,x} & c_{y,y} & c_{y,z} \\ c_{z,x} & c_{z,y} & c_{z,z} \end{bmatrix} \quad (6)$$

Where $c_{x,y}$ is the covariance between x and y shown in (7) and so do the others.

$$c_{x,y} = \text{cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1} \quad (7)$$

By using eigenvalue decomposition in (8), the feature vectors can be obtained.

$$C = Q\Sigma Q^{-1} \quad (8)$$

Where Q is the eigenvector matrix, Σ is a diagonal matrix whose diagonal element corresponds to the eigenvalue. The directions of the first two feature vectors with the largest eigenvalues corresponds to directions of the length and width of the leader vehicle. For calculation easily, point clouds are transformed to a new coordinate system defined as F_Q whose axis corresponds to eigenvectors. The transformation is shown in (9).

$$P_c^Q = Q^{-1} \left(P_c - (\bar{x}, \bar{y}, \bar{z})^T \right) \quad (9)$$

Where P_c^Q represents point cloud after transformation.

We calculate the length information respectively along x and y axis in F_Q defined as $L1$ and $L2$. Due to the random placement of the leader, it cannot be judge whether $L1$ or $L2$ corresponds to the width of the vehicle. In this paper, an off-road vehicle is selected as the leader with 1.8 meters in width defined as W_{ref} . Based on this, $L1$ and $L2$ with the smallest difference from W_{ref} is defined as the width defined as W_{tar} and the other is the length of the vehicle defined as L_{tar} . Meanwhile boundary points can be easily obtained in F_Q and they are defined as $B^Q = (x_{\min}^Q, y_{\min}^Q, z_{\min}^Q, x_{\max}^Q, y_{\max}^Q, z_{\max}^Q)$. Fig.6(a) shows the minimum bounding box of each clustering block. According to B^Q , we can obtain the geometric center of each block defined as C_g^Q shown in (10).

$$C_g^Q = \left(\frac{x_{\min}^Q + x_{\max}^Q}{2}, \frac{y_{\min}^Q + y_{\max}^Q}{2}, \frac{z_{\min}^Q + z_{\max}^Q}{2} \right) \quad (10)$$

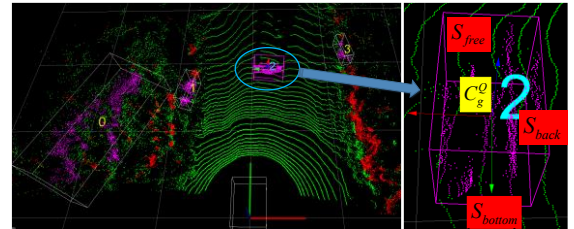


Fig.6 Geometric model based vehicle detection.

According to the distribution characteristics of point clouds scanned to the vehicle, point clouds in each cluster block can be divided into three parts as shown in Fig.6(b). With reference to C_g^Q , the back part is defined as S_{back} where most of point clouds are distributed in this part. For the rest, the part below C_g^Q is defined as S_{bottom} and the other is defined as S_{free} which has the fewest point clouds. Count the amount of each part defined as C_{back} , C_{bottom} and C_{free} . The statistics data are used as a condition for vehicle model judgment as shown in (11). It is effective for distinguishing clustering blocks evenly distributed such as bushes or trees.

$$\begin{cases} C_{back} > 2 \cdot C_{free} \\ C_{bottom} > C_{free} \\ C_{back} + C_{bottom} + C_{free} > 100 \end{cases} \quad (11)$$

The height of each block H_{tar} is defined as the maximum height difference from the ground shown in (12).

$$H_{tar} = \max \left\{ \|z_i - G^{ref}\| \mid z_i \in Z_{tar} \right\} \quad (12)$$

Where G^{ref} is the reference height of the surrounding ground plane similar to $G_{x,y}^{ref}$ and $z_i \in Z_{tar}$ is the height value of point cloud in each block. The size of each block needs to meet geometric constraints based on the actual vehicle model shown in (13).

$$\begin{cases} 0.5 < W_{tar} < 2.0 \\ L_{tar} < 6.0 \\ 1.4 < H_{tar} < 2.5 \end{cases} \quad (13)$$

The proposed method is mainly used for vehicle tracking, so for multi-targets that satisfy constraints, the one with the minimum lateral deviation value from the follower is selected as the target. And the centroid of the target block is used as the target location shown in (14).

$$P_{tar} = \arg \min_{C_i(\bar{x}, \bar{y}, \bar{z})} \|\bar{x}_i\| \quad (14)$$

D. Detection based Auxiliary Vehicle Tracking

The detection result can also be used for continuous tracking in close to make up the blind area of MMW Radar during turning. Based on the historical trajectory of the leader, we use Kalman predictor to obtain the predicted position at the next moment.

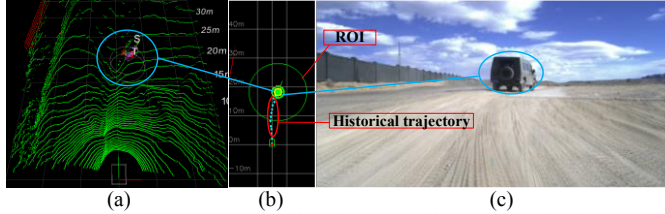


Fig. 7 Analysis of movement trend based on 3D-Lidar. (a) point cloud space (b) 2D grid map (c) image.

As shown in Fig.7(b), the cyan dotted line represents the historical trajectory and the yellow-green solid circle represents the current target. This dealing can be used to extract the region of interest represented by the green circle in Fig.7(b) during the tracking. And the speed of the leader can also be derived by Kalman predictor.

Due to the echo movement of the follower, it is difficult to effectively analyze the motion state of the leader in F_v . So we transform $P_{tar}^v = (x^v, y^v, z^v)$ to $P_{tar}^G = (x^G, y^G, z^G)$ as shown in (15) to achieve the analysis of the leader's movement trend.

$$\begin{cases} \begin{bmatrix} x^G \\ y^G \end{bmatrix} = \begin{bmatrix} vx^G \\ vy^G \end{bmatrix} + \begin{bmatrix} \sin \alpha & \cos \alpha \\ -\cos \alpha & \sin \alpha \end{bmatrix} \cdot \begin{bmatrix} x^v \\ y^v \end{bmatrix} \\ z^G = z^v \end{cases} \quad (15)$$

Where (vx^G, vy^G) is the echo global position of the follower in F_G and α is the current yaw angle. $z_t = [x^G \ y^G]^T$ is

defined as the observation matrix at time t , $x_t = [px^G \ py^G \ pv_x^G \ pv_y^G]^T$ is defined as the state matrix, (px^G, py^G) is the predicted global position of the leader at $t+1$, and (pv_x^G, pv_y^G) is the predicted speed component. According to Kalman predictor, the prediction information can be obtained by (16).

$$x_{t+1} = x_t + K_t \cdot (z_t - H \cdot x_t) \quad (16)$$

Where K_t is the gain coefficient and $H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$.

According to (px^G, py^G) , (px^v, py^v) can be calculated by using the inverse transformation of (15) and it can be used to extract the region of interest in the next frame. During tracking, for multiple candidate detection results, the one closest to (px^v, py^v) is selected as the tracking target.

IV. MMW RADAR BASED VEHICLE TRACKING

MMW Radar is the main sensor used for target tracking. It is an ideal moving target tracking sensor as it can directly obtain the relative speed of the target. The relative speed is of great use to eliminate static obstacles during tracking. Fig.8(a) shows the scene when the leader is static and Fig.8(b) shows the moving scene. The grey circle means static obstacle whose relative speed is zero and the cyan circle means dynamic obstacle whose relative speed is greater than zero. For the vehicle tracking process based on MMW Radar, the main work is to keep tracking the same target and eliminate the interference of other dynamic obstacles.



Fig. 8 MMW Radar based vehicle tracking.

After the target of the leader is detected for tracking target initialization, the MMW Radar based tracking starts. Based on the position of detected target from 3D-Lidar, the dynamic target closest to the initialization result is regarded as the tracking target by MMW Radar.



(a) 2D grid map (b) image
Fig. 9 Analysis of movement trend based on MMW Radar.

The same as last section, we record the historical tracking result based on MMW Radar for a period to form a trajectory represented by the yellow dotted line in Fig.9(a). The trajectory is also transformed to F_G for analysis of movement trend by Kalman predictor. And the predicted position is also used for extraction of region of interest in the next frame represented by the green circle in Fig.9(a).

For each tracking result of MMW Radar, it can automatically track the target with the same index when the target can be detected by the sensor. In the continuous tracking processing, according to the last detection result, if the target with the same index is still valid and its position is within ROI, the current tracking is valid.

However, due to the disturbances such as driving on the bumpy road, turning a corner and occlusion, the continuous tracking with the same index will be interrupted in unstructured environment. Sometimes the target may be detected by the sensor in other detecting direction with a different index. In this case, if the candidate closest to the predicted position is within ROI and its speed is consistent with the predicted speed, it is re-selected as the tracking target. Then update the index of continuous tracking target and go on.

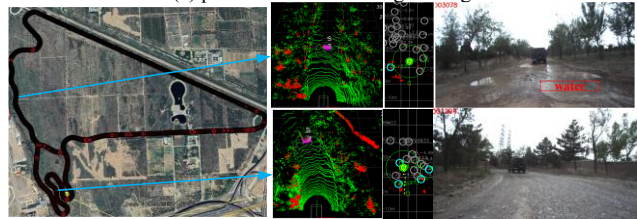
For the loss of tracking based on MMW Radar, it switches to auxiliary tracking based on 3D-Lidar as described above. This situation often occurs in corners as its horizontal field of view is narrow. When the target is re-tracked by MMW Radar, tracking based on MMW Radar is switched to continue.

V. EXPERIMENTS

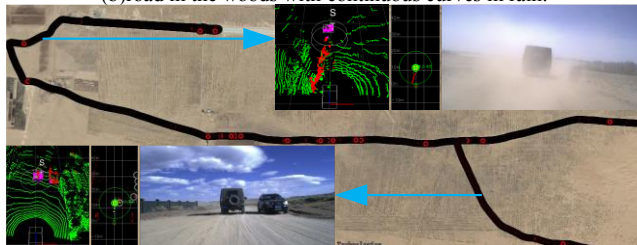
We do a lot of experiments in various terrain environment including forest roads with partial occlusion, sand roads with dust and bumpy roads. The total driving mileage reaches hundreds of kilometers.



(a) partial occlusion during turning.



(b) road in the woods with continuous curves in rain.



(c) road in the desert.

Fig.10 Typical unstructured road environment.

Fig.10 shows tracking scenes under some typical harsh conditions. Fig.10(a) shows partial occlusion by the building when the leader turns. This situation has almost no effect on our tracking. Since we record the historical trajectory, even if the target disappears temporarily, the continuity of tracking can still be guaranteed through target re-detection after the follower drive along the historical trajectory to the location where the leader disappears. Fig.10(b) shows roads in the woods with continuous curves. We did a special experiment in rain and the weather has no effect on our tracking method. Fig.10(c) shows roads in the desert. There is a lot of dust during driving especially in windy weather. Dust and water will cause interference to tracking based on 3D-Lidar or camera, but they have no effect for MMR Radar. Besides, for vehicle interaction scenes, according to the movement trend based on Kalman predictor, it can effectively distinguish the interference of other vehicles, as they drive outside the predicted ROI of the leader.

We choose a bad test environment with heavy dust, slopes and sharp turning shown in Fig.10(c) to compare the proposed method with those by single sensor. Table I shows the comparison of statistical result during actual tracking process among the proposed method, tracking by 3D-Lidar and tracking by MMV Radar.

TABLE I. COMPARISON OF STATISTICAL RESULT

Method	Total frames	Lost frames	Precision rate
Proposed	15784	19	99.88%
3D-Lidar	15784	1590	89.93%
MMV Radar	15784	512	96.76%

Heavy dust and long-distance tracking are main reasons for tracking failure by 3D-Lidar. Target re-initialization after vehicle stopping and sharp turning are main reasons for tracking failure by MMV Radar. Tracking failure by the proposed method only happens during the uphill process, but maintenance of historical target trajectory can still ensure the leader-follower formation transportation is not affected. And it can automatically complete the re-initialization of tracking target.

In addition, the distance of stable tracking can reach 50 meters at speed of 40km/h to ensure safety in emergency braking situations. And the average speed keeps about 20km/h in general unstructured environment. Different speed almost has no effect on the continuous tracking even with heavy dust.

VI. CONCLUSION

The proposed method is stable and real-time. It overcomes the limitations of tracking by single sensor and take advantage of multi-sensor fusion to implement the initial detection and continuous tracking of the leader. The average processing time is within 50ms. Since the sensors used in this paper does not depend on light, the proposed method can be applied day and night compared with that based on camera. It is suitable for long-distance unmanned formation. In subsequent research, we will use technical means to achieve cost reduction. For example, the echo pose information can be obtained by 3D-Lidar based SLAM technology, which will replace GPS/IMU. This is of great significance to promote the development of unmanned formation technology.

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